

Amit Kumar Mohanty

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Target Role: AI Solution Architect I | HCLTech

PROFESSIONAL SUMMARY

Generative AI Engineer with 1+ year of hands-on experience designing end-to-end GenAI and Agentic AI solutions - including Retrieval-Augmented Generation (RAG) pipelines, multi-agent workflows, and LLM integration layers. Strong foundation in system design (microservices, REST APIs, event-driven and cloud-native / 12-factor patterns) applied to production-grade, scalable AI services. Skilled in LLM evaluation, prompt engineering, LLMOps (tracing, monitoring, feedback loops), and Responsible AI (fairness, bias mitigation, prompt-injection safety). Seeking to grow into an AI architecture role, translating business goals into secure, production-ready GenAI systems on Azure, AWS, and GCP.

EDUCATION

K.R. Mangalam University, Gurgaon July 2021 - July 2025

B.Tech in Computer Science and Engineering (Specialization in AI/ML)

Birla Institute of Technology and Science, Pilani (BITS Pilani) July 2026 - July 2028

M.Tech in Artificial Intelligence and Machine Learning (Pursuing)

EXPERIENCE

Research Associate - AI, Keywords Studios India - Gurgaon, Haryana May 2025 - Present

- Designed and implemented Agentic AI workflows - agent orchestration, tool use, and multi-step long-horizon task execution - improving task-completion reliability.
- Contributed to reference patterns and quality guardrails for GenAI agents, running structured design/test reviews with a global team to deliver client-ready iterations.
- Architected data pipelines (ingestion, preprocessing, feature engineering, versioning) that fed 85% of self-generated training datasets to production; awarded "Rookie of the Year 2025."

Artificial Intelligence Intern, Infosys - Remote Nov 2024 - Feb 2025

- Designed an LLM integration layer and prompt-management approach for an AI call-center system with LangChain/LangSmith, raising response accuracy by 35%.
- Built an LLM evaluation and observability loop (automated tracing + scoring) that reduced hallucinations by 20% and mitigated biased/undesired outputs.
- Optimized LLM deployment for latency (-25%) within cloud-native services, collaborating across a 12-engineer team using Agile/Scrum.

PROJECTS

Dynamic Screen Companion - GenAI Solution Architecture June 2025 - Present

- Architected an end-to-end multi-modal GenAI system: LLM integration layer, RAG retrieval, and a FastAPI microservice backend handling concurrent inference streams.
- Designed the embedding + vector-search data layer (chunking, embeddings, retrieval) and containerized the solution (Docker, CI/CD) for scalable cloud deployment.

Diabetic Optiscan - Vision Transformer (Peer-reviewed research) Jan 2025 - June 2025

- Designed and led integration of a Vision Transformer (ViT-B/16) on the APTOS dataset; co-authored a peer-reviewed paper on the architecture and accuracy gains.
- Created a custom wavelet technique that improved detection precision by 50% over legacy methods.

SKILLS

AI Architecture & System Design: Solution Design, Microservices, Event-Driven Architecture, REST API Design, Cloud-Native / 12-Factor, Distributed Systems, Scalability, Reliability & Latency Optimization

GenAI & Agentic AI: LLMs, RAG, Multi-Agent Systems, Agent Orchestration, Tool Use, Memory Management, Human-in-the-Loop, Prompt Engineering, Chain-of-Thought, LLM Integration

Frameworks: LangChain, LangGraph, LlamaIndex, Hugging Face, Transformers, PyTorch, TensorFlow, FastAPI

Fine-tuning & LLMOps: LoRA, QLoRA, PEFT, Prompt Tuning, MLflow, Model Evaluation Frameworks, Tracing, Monitoring, Feedback Loops, Docker, Kubernetes, CI/CD

Cloud, Data & Vector Stores: Azure OpenAI, AWS, GCP (Vertex AI), Vector Databases (FAISS, Chroma, Pinecone), Embedding Stores, Data Pipelines, SQL, MongoDB

Responsible AI & Security: AI Fairness, Explainability, Bias Mitigation, Prompt-Injection Protection, Data Privacy (GDPR), Audit Logging

CERTIFICATIONS

- Ducat India - Data Science Professional Training (Python, ML, statistics, predictive modeling), Apr 2024 - Apr 2025.
- ISRO - AI/ML for Geodata Analysis (applied ML/DL to remote-sensing imagery).
- Actively pursuing cloud & AI architecture certifications (Microsoft Azure AI, AWS, Google Cloud).